

Periods and L -functions of meromorphic modular forms via finite contours

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Abstract

In this note, we assign to every meromorphic modular form f of weight k on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ a finite L-integral, which we denote $f \mapsto L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$; here τ_0 is a base-point in the upper half-plane, and $\gamma_{\tau_0}^S$ is an integration path from $-\frac{1}{\tau_0}$ to τ_0 , while $\gamma_{\tau_0}^T$ an integration path from $\tau_0 - 1$ to τ_0 . These L-integrals are finite for all base-points and contours. For the special case of weakly holomorphic modular forms, i.e. where f is either bounded at the cusp or has poles exclusively at the cusp, this L-integral is uniquely defined, i.e. it is fully independent of τ_0 and is fully independent of $\gamma_S(\tau_0)$ and $\gamma_T(\tau_0)$. We prove that this finite L-integral, as we define it, exactly matches the L-integral of holomorphic modular forms, and, separately, that it equals the non-holomorphically regularized L-integrals of [1]. When f has poles away from the cusp, i.e. inside the fundamental domain, $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ is unique and constant for paths within the same homotopy class; related L^* -integrals of f , evaluated at s , which depend on paths outside the same homotopy class differ by well-defined wall-crossing residues. We prove that, for meromorphic modular forms regular at the cusps, our L-integrals/L-functions equal the L-integrals/L-functions constructed in [2]. Finally, we show that a slight modification of this linear functional in the Fourier coefficients of f yields its associated period polynomial when k is such that $\dim(S_k) = 1$. Our period polynomials exactly reproduce the numerical periods reported in [3]. All results in formulas are verified, and are verifiable, via our python scripts [4].

1 Introduction and main results

In this paper, we introduce a general framework that defines L-functions for generic modular forms on the full group $\mathrm{SL}_2(\mathbb{Z})$, and extracts their periods for the explicit case when the holomorphic cusp forms are one-dimensional, i.e. when $\dim(S_k) = 1$. Classically, L-functions and periods are both well-defined and well-studied for holomorphic modular forms. However, there is a growing body of work exploring L-functions [1] and periods [3] for weakly holomorphic modular forms. Similarly, there are natural physical motivations to consider L-functions for properly meromorphic modular forms, which have poles *within* the fundamental domain [2].

In these works, and specifically in [1, 2], and throughout physics, divergences due to singularities in quantities of interest necessitate the introduction of regulators in order to define finite quantities. Regulators, though, often break symmetries, and care is needed to check that the regulator has not injected undue violence to the system. Of relevance here is that [1] introduced a non-holomorphic regulator to deal with the singularities at either end of the classical Mellin contour, $\int_0^\infty$, sufficient to regulate divergences at the cusps, but which violates holomorphicity in the deformed contour-integral, ruining path-dependence. Simultaneously, [2] concerned itself with L-functions for modular forms which diverge within the interior of the fundamental domain. Here, holomorphic dependence on τ is paramount, which puts this treatment at odds with the regulator in [1].

In this short note, we introduce manifestly finite contour integral functionals to define L-functions for generic meromorphic modular forms that are of sufficient generality to handle any

combination of poles at cusps and poles within the fundamental domain. Our construction is motivated by, and makes precise contact with, work which extracts period-coefficients from weak cusp forms [3]. Our integrals are manifestly finite throughout, and reproduce all results in the literature, unifying the quasi-period results and the L-function extensions within one concrete, natural framework.

What we construct. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, let f be a meromorphic modular form of even weight $k \in \mathbb{Z}$ on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint with the property that the pair of paths

$$\begin{aligned} S\text{-segment: } & \gamma_{\tau_0}^S : [0, 1] \rightarrow \mathbb{H}, \quad \gamma_{\tau_0}^S(0) = -1/\tau_0, \quad \gamma_{\tau_0}^S(1) = \tau_0, \\ T\text{-segment: } & \gamma_{\tau_0}^T : [0, 1] \rightarrow \mathbb{H}, \quad \gamma_{\tau_0}^T(0) = \tau_0 - 1, \quad \gamma_{\tau_0}^T(1) = \tau_0, \end{aligned} \quad (1)$$

in $\mathbb{H}$ avoids the Γ -orbit of the poles of f in $\mathbb{H}$. Using these contours, we define

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right], \quad (2)$$

where the T -kernel is built from the following explicit two Hurwitz-zeta functions:

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1). \quad (3)$$

When the contour avoids poles in f , both contour integrals contributing to $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ in (2) are finite for every basepoint $\tau_0 \in \mathbb{H}$ admissible for f and every $s \in \mathbb{C}$; no regulator, asymptotic-subtraction argument, or analytic-continuation step is required. Further, as we will show, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is invariant under both τ_0 -variations and $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ -variations, as long as neither $\gamma_{\tau_0}^S$ nor $\gamma_{\tau_0}^T$ cross any pole location within the fundamental domain.

We will show that $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is the classical L-function for $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Further, because of its τ_0 -independence and finiteness for $f \in M_k^!(\mathrm{SL}_2(\mathbb{Z}))$, and because we reproduce the analytic formulae in [1], we unambiguously show that their candidate L-function is the canonical L-function one would associate with any weak modular form, $f \in M_k^!$.

Three theorems. The functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ obeys three properties that together identify it as the canonical finite-contour L-function of f :

- (i) (*Theorem 1.1*) When $\dim S_k = 1$, the integer- s specialisations of (2) reduce to a polynomial-kernel form that produces the periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$ as an explicit linear functional in the Fourier coefficients of f . For $f = \Delta$ at $k = 12$ this recovers the Brown values of $\omega^\pm(\Delta)$; for $f = \hat{\Delta}$ this gives the quasi-periods $\eta^\pm(\hat{\Delta})$ from a finite-endpoint contour pair, with no divergence at the cusp.
- (ii) (*Theorem 1.2*) For $f \in M_k^!$, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is τ_0 -independent on $\mathbb{H}$ and produces the completed L-function $L^*(f, s)$ at every $s \in \mathbb{C}$ as a single closed-form incomplete-gamma sum, agreeing term by term in the Fourier expansion with [1]. Our proof in fact exploits the simplicity at the elliptic fixed-point basepoint $\tau_0 = i$, where the S -segment collapses to a point and (2) reduces to a single integral on the compact contour $[i-1, i]$.
- (iii) (*Theorem 1.3*) For general meromorphic f , $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ only through their homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, with explicit wall-crossing residue jumps when the homotopy class changes by crossing a pole. Restricted to forms regular at the cusp, this reproduces the deformed-Mellin L-function $L_f^*(s)_{\mathrm{int}}$ (Corollary 1.4; see [2]).

The complete statements are below; their proofs occupy §3–§5.

Theorem 1.1 (Periods and quasi-periods via the integer- s specialisation of L^* , $\dim S_k \leq 1$). *Let $k \in 2\mathbb{Z}$ with $\dim S_k \leq 1$, and let $f \in S_k^!$. For the substantive cases $k \in \{12, 16, 18, 20, 22, 26\}$ where $\dim S_k = 1$, the period and quasi-period coefficients $\omega^\pm(f)$ are recovered as the integer- $s = j$ specialisation of the L -integral functional of Theorem 1.2 via*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{j-1} \binom{k-2}{j-1}}{c_S^{j-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{j-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, j) d\tau \right] \quad (4)$$

for every $f \in S_k^!$, every $\tau_0 \in \mathbb{H}$, and every interior $j \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $j-1 \bmod 2$. Here $c_S^{j-1} := [p_f^\pm]_{X^{n-(j-1)} Y^{j-1}} \in \mathbb{Q}$ is the Brown cuspidal-basis coefficient at the matched monomial ([3, 13]), and $k_T(\tau, s)$ is the Hurwitz-zeta T -kernel of (3), whose integer- $s = j$ specialisation collapses via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$) to the $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, j) = -\frac{B_j(\tau+1)}{j} + (-1)^{j-1} \frac{B_{k-j}(\tau+1)}{k-j} \in \mathbb{Q}[\tau] \quad (5)$$

of degree at most $k-2$. The right side of (4) is independent of τ_0 and of the choice of interior j within its parity sector (lockstep). The S - and T -segment kernels are the bare Mellin and bare Hurwitz kernels of $L^*(f, s)$: at integer $s = j$, identity (4) is the rescaled integer- s shadow of the L -function functional of Theorem 1.2, with all combinatorial structure absorbed into the prefactor outside the two-segment bracket.

Higher-dimensional S_k . For $k \geq 24$ with $\dim S_k \geq 2$, the Brown cuspidal-basis coefficient c_S^{j-1} in (4) becomes matrix-valued: each $j \in \{2, \dots, k-2\}$ gives a $\dim S_k$ -component linear functional on $\{c_f(n)\}$ rather than a scalar. The structural argument behind Theorem 1.1 — coboundary inversion on V_{k-2} via δ_T^{-1} in closed Bernoulli-operator form — generalises without conceptual change, but the explicit kernel tables grow with $\dim S_k$, and the lockstep / basepoint-independence statement becomes a matrix identity. Working out the explicit matrix-valued kernels for $k = 24, 28, 30, \dots$ and connecting them to the existing Eichler–Shimura matrix-period basis literature is the natural generalisation of Theorem 1.1, and we leave it to a follow-up. Theorems 1.2 and 1.3 are unaffected and apply at every even k .

Theorem 1.2 (L -function for weakly holomorphic forms). *Let $k \in 2\mathbb{Z}$ and $f \in M_k^!$. With $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3), for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (6)$$

is finite and independent of both $\tau_0 \in \mathbb{H}$ and of the admissible contour choices $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$. We write $L^*(f, s)$ for the common value.

Specialising to the elliptic fixed point $\tau_0 = i$ of the modular involution $S: \tau \mapsto -1/\tau$, the S -segment $[-1/i, i]$ collapses to the single point i , the S -integral in (6) vanishes, and L^* reduces to a single integral on the compact contour $[i-1, i] \subset \mathbb{H}$, with closed-form sum

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (7)$$

The right side of (7) is the completed L -function $L^*(f, s)$ of f : on holomorphic cusp forms it reduces to the classical Hecke–Weil Mellin transform, and on $M_k^!$ its equality to the formula of [1] proves that the incomplete-gamma construction — previously obtained via non-holomorphic regulator deformations — is the unique canonical L -function for every weakly holomorphic modular form.

Classical Mellin recovery and canonical uniqueness. For $f \in S_k$ a holomorphic cusp form, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in (6) collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while the T -segment vanishes exponentially (since $f(\tau) = O(e^{-2\pi \operatorname{Im} \tau})$ on the segment as $\operatorname{Im} \tau \rightarrow \infty$):

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt, \quad (8)$$

the classical completed L -integral of f . For $f \in M_k$ with $c_f(0) \neq 0$ the same limit holds after a standard constant-term subtraction (see [2, Lemma 2.1] for the explicit polynomial correction).

For $f \in M_k^!$ with cusp poles the cusp-limit is genuinely pathological: $f(\tau) \rightarrow \infty$ as $\operatorname{Im} \tau \rightarrow \infty$, so the S -segment integral diverges along the imaginary axis, and the T -segment integral $\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau$ acquires a matching divergence of the opposite sign as $\operatorname{Im} \tau_0 \rightarrow \infty$. The two divergences cancel at every finite τ_0 , so $L^*(f, s)$ is finite and well defined throughout $\mathbb{H}$, but no naive Mellin contour limit exists. The construction of [1] recovers $L^*(f, s)$ on $M_k^!$ by a non-holomorphic regulator at the cusps; Theorem 1.2 shows that this regulator and the finite-contour functional (6) compute the same object, term by term in the Fourier expansion. In particular, $L^*(f, s)$ is independent of any choice of regularisation: it is the value of the finite-contour functional at any admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$, and the BFK construction is one closed-form evaluation of it (at $\tau_0 = i$).

Theorem 1.3 (*L -function for meromorphic modular forms*). *Let $k \in 2\mathbb{Z}$ and let f be a meromorphic modular form of weight k on Γ . Write $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ for the upper half-plane with the Γ -orbit of the interior poles of f removed, and call the pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ admissible if $\tau_0 \in \mathbb{H}_f$ and both contours lie in $\mathbb{H}_f$, with $\gamma_{\tau_0}^S$ running from $S^{-1}\tau_0 = -1/\tau_0$ to τ_0 and $\gamma_{\tau_0}^T$ from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 . Define $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ by (2) with the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3). Then:*

- (a) *Both integrals in (2) are finite for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and every $s \in \mathbb{C}$.*
- (b) *$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on the admissible pair only through its homotopy class in $\mathbb{H}_f$: any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint τ_0 — leaves the value unchanged.*
- (c) *If two admissible configurations $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ lie in homotopy classes related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f , then*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^S) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^S) = \mathcal{R}_{\tau_p}(f, s), \quad (9)$$

where for a simple pole $\operatorname{Res}(f, \tau_p)$,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (10)$$

with $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ the winding-number contributions of $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ across τ_p (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in τ_p of (10); the closed form is given in §5.

Corollary 1.4 (L^* on regular-at-cusps meromorphic forms). *Let $f \in F_k$ be a meromorphic modular form of weight k that is regular at the cusp. Let $L_f^*(s)_{\text{int}, \gamma}$ denote the deformed-Mellin L -function of f along a contour γ from 0 to $i\infty$ lying in $\mathbb{H}_f$ (see [2, §2]). Then for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ whose homotopy class in $\mathbb{H}_f$ is compatible with γ (in the sense that the contour $\gamma_{\tau_0}^S$ together with the cusp-limit piece $\lim_{\tau_0 \rightarrow i\infty} \gamma_{\tau_0}^S \rightarrow [0, i\infty]$ can be deformed to γ in $\mathbb{H}_f$),*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_f^*(s)_{\text{int}, \gamma} \quad \text{for every } s \in \mathbb{C}. \quad (11)$$

The structural picture. The contour pair (1) has two natural one-parameter specialisations of the basepoint, each exploiting a fixed point of the modular curve $Y(1)$: the cusp specialisation $\tau_0 \rightarrow i\infty$ sends the S -segment to the classical Mellin contour $[0, i\infty]$ and the T -segment to 0 exponentially; the elliptic specialisation $\tau_0 = i$ kills the S -segment identically and leaves L^* as a single integral on $[i-1, i]$. The cusp specialisation, applied at finite $\tau_0 = i\Lambda$, gives the Λ -truncated Mellin contour of $L_f^*(s)_{\text{int}}$ (see [2]) as the S -segment alone, with the T -segment serving as the missing ingredient that produces the cocycle and renders the $\Lambda \rightarrow \infty$ limit unnecessary. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ in (3) is the closed form of the T -segment kernel: it is the unique V_n -analog, valid at every $s \in \mathbb{C}$, of the polynomial coboundary inversion of [7] (which itself recovers as the integer- s specialisation, by Theorem 1.1).

Outline. §2 introduces the period-defining cocycle and its moment expansion in the polynomial coefficient system V_{k-2} . §3 identifies the bare Mellin and Hurwitz integer- s specialisations as the period-extracting kernels and proves Theorem 1.1. §4 derives the Hurwitz-zeta kernel $\tilde{k}_T$ via a coboundary equation, proves Theorem 1.2 via a Hurwitz-unfolding lemma, and contains an explicit worked example at $k = 12$ exhibiting the Bernoulli-polynomial collapse of $\tilde{k}_T$ at integer s together with numerical values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$. §5 proves Theorem 1.3 and Corollary 1.4. §6 collects conceptual framings (holonomy/Wilson-loop interpretation, on-shell-amplitudes analogy, the regularisation-as-symptom heuristic, the open question on the geometric content of the $\tau^{s-1} f d\tau$ Mellin kernel, and pointers to the sequel on systematic per-mode theory for interior-pole forms). Appendix A records the algebraic derivation of the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ from contour-invariance of the two-segment functional: starting from the classical Mellin transform split at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding that the resulting incomplete-gamma tail be representable as a T -segment integral, one derives a shift-difference equation whose solution is the Hurwitz-zeta combination of (3). Appendix B is a brief Eichler–Shimura primer for readers approaching the subject from theoretical physics. This appendix is included for the working-version of this note and is intended to be removed before formal publication.

2 The period-defining cocycle and its moment expansion

The polynomial coefficient system. Let $V_n := \mathbb{C}[X, Y]_n$ be the space of complex homogeneous polynomials in two auxiliary variables X, Y of degree n , equipped with the right Γ -action

$$(P|_\gamma)(X, Y) = P(aX + bY, cX + dY) \quad \text{for } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

For a meromorphic modular form f of weight k , write $n := k-2$ throughout and form the V_n -valued 1-form on $\mathbb{H}$

$$\omega_f(\tau) := f(\tau) (X - \tau Y)^n d\tau.$$

Modularity of f implies that ω_f is Γ -equivariant in the standard sense [13]: under $\tau \mapsto \gamma^{-1}\tau$, the weight- k transformation of f cancels the weight- $(-n)$ slash on $(X - \tau Y)^n$, so the V_n -valued 1-form ω_f is invariant under the simultaneous action of Γ on τ and on the auxiliary variables (X, Y) . Equivalently, ω_f is a well-defined V_n -valued 1-form on the modular curve $Y(1) = \Gamma \backslash \mathbb{H}$ (modulo the Γ -action on V_n).

The cocycle. The two-segment cocycle construction used in the present note originates in personal correspondence from F. Brown to the author [5], which describes the procedure of integrating a weakly holomorphic modular form against the polynomial coefficient system $(X - \tau Y)^n$ along contours from $\gamma^{-1}\tau_0$ to τ_0 for the generators $\gamma \in \{S, T\}$ of Γ , and extracting periods and quasi-periods from the resulting 1-cocycle modulo coboundaries. Concretely: fix a basepoint $\tau_0 \in \mathbb{H}$ not on the Γ -orbit of any pole of f , and define

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f = (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}, \quad (12)$$

the contour being any path from $\gamma^{-1}\tau_0$ to τ_0 in $\mathbb{H}$ avoiding the Γ -orbit of the poles of f . The integral is independent of the choice of path within the same homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Lemma 2.1 (Cocycle property). *For all $g, h \in \Gamma$,*

$$C_{gh}^f(\tau_0) = C_g^f(\tau_0)|_h + C_h^f(\tau_0). \quad (13)$$

Equivalently, the map $\gamma \mapsto C_\gamma^f(\tau_0)$ is a 1-cocycle of Γ with coefficients in $V_n \otimes \mathbb{C}$.

Proof. Path-additivity gives

$$\int_{(gh)^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} + \int_{h^{-1}\tau_0}^{\tau_0}.$$

In the first integral substitute $\tau = h^{-1}\sigma$, so σ runs from $g^{-1}\tau_0$ to τ_0 . The Γ -equivariance of ω_f converts the integrand to $\omega_f(\sigma)|_h$, yielding the first summand of (13). The second integral gives the second summand directly. $\square$

A cocycle on $\Gamma = \text{SL}_2(\mathbb{Z})$ is determined by its values on the two generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

subject to the relations $S^2 = (ST)^3 = -I$ (which act trivially on V_n for n even). We write

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \int_{-1/\tau_0}^{\tau_0} \omega_f, \quad C_T^f(\tau_0) = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} \omega_f, \quad (14)$$

since $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$. These are the two integrals over the contour pair (1).

Moment expansion. Expanding $(X - \tau Y)^n = \sum_{j=0}^n \binom{n}{j} (-\tau)^j X^{n-j} Y^j$ and pulling the polynomial out of the integral defines the moment integrals

$$I_S^j(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau, \quad I_T^j(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^j d\tau, \quad (15)$$

for $j = 0, 1, \dots, n$, in terms of which

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{j=0}^n \binom{n}{j} (-1)^j I_S^j(f, \tau_0) X^{n-j} Y^j,$$

and similarly for $C_T^f(\tau_0)$. Theorem 1.1 extracts $\omega^\pm(f)$ from particular linear combinations of these moments.

Basepoint dependence. Varying τ_0 changes C^f by a coboundary, provided f has no interior poles. Define a primitive $F(\tau)$ of ω_f on a simply connected open subset of $\mathbb{H}$ avoiding $\Gamma \cdot f^{-1}(\infty)$. Then for any two basepoints τ_0, τ'_0 in the same connected component, path-additivity gives

$$C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = (P|_\gamma - P) \quad \text{with} \quad P = (2\pi i)^{n+1} (F(\tau'_0) - F(\tau_0)) \in V_n \otimes \mathbb{C}. \quad (16)$$

This is the coboundary $\delta P(\gamma) := P|_\gamma - P$. When f has interior poles, dragging τ_0 across a pole picks up a Cauchy residue contribution beyond (16); this wall-crossing structure is the content of Theorem 1.3 in §5.

Cohomology and periods. The cohomology class $[C^f] \in H^1(\Gamma; V_n)$ is independent of τ_0 (on the component) and independent of the choice of δP representative. The classical Eichler–Shimura theorem [13] identifies the cuspidal subspace $H_{\text{cusp}}^1(\Gamma; V_n)$ with $S_k \oplus \overline{S_k}$, and the linear coordinates of $[C^f]$ in this cuspidal subspace are the periods and quasi-periods $\omega^\pm(f), \eta^\pm(f)$ [3].

3 Period extraction: proof of Theorem 1.1

Set $n := k-2$ throughout. The argument is k -uniform on the $\dim S_k = 1$ family $k \in \{12, 16, 18, 20, 22, 26\}$: every step below — the cocycle structure of §2, the δ_T -solvability of Lemma 3.1, the Bernoulli closed-form inversion of Proposition 3.2, the Hurwitz-zeta kernel identification of Proposition 3.3, and the lockstep of Corollary 3.4 — holds for every even k with the Brown cuspidal-basis coefficient c_S^{j-1} replaced by the relevant weight's basis-polynomial coefficient. The lower even weights $k \in \{2, 4, 6, 8, 10, 14\}$ in the $\dim S_k \leq 1$ family have $S_k = 0$, so the theorem is vacuous there. Numerical verification at $k = 12$ for the canonical pair $f \in \{\Delta, \widehat{\Delta}\}$, where Brown's published values $\omega^\pm(\Delta), \eta^\pm(\widehat{\Delta})$ [3] provide a 50-digit benchmark, is carried out in the companion script `verify_periods.py` [4]; analogous numerical verification at $k \in \{16, 18, 20, 22, 26\}$ requires only the Brown basis coefficients c_S^{j-1} at those weights and is structurally identical.

Lemma 3.1 (Solvability of $\delta_T P = C_T^f$ is cuspidality of f). *Define the linear endomorphism $\delta_T: V_n \rightarrow V_n$ by $\delta_T(P) := P|_T - P$, where $T \in \Gamma$ acts on V_n as $(X, Y) \mapsto (X + Y, Y)$. Then:*

- (a) $\ker(\delta_T) = \mathbb{C} \cdot Y^n$.
- (b) $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$.
- (c) *For any meromorphic modular form f of weight $k = n + 2$ and any $\tau_0 \in \mathbb{H}$ admissible for f ,*

$$[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} \cdot c_f(0), \quad (17)$$

where $c_f(0)$ is the constant Fourier coefficient of f .

Consequently, the equation $\delta_T P = C_T^f(\tau_0)$ is solvable for $P \in V_n$ if and only if $c_f(0) = 0$, i.e. if and only if f is a (weak) cusp form. When solvable, P is unique modulo $\mathbb{C} \cdot Y^n$.

Proof. (a) T acts on V_n as the translation $X \mapsto X + Y$, fixing only powers of Y . Hence T -invariants in V_n , equivalently $\ker(\delta_T)$, are exactly $\mathbb{C} \cdot Y^n$.

(b) Writing $P = \sum_{j=0}^n p_j X^{n-j} Y^j$ and expanding $(X + Y)^{n-j}$ binomially,

$$[\delta_T P]_{X^{n-m} Y^m} = \sum_{j=0}^{m-1} \binom{n-j}{m-j} p_j \quad (m = 0, 1, \dots, n). \quad (18)$$

This is strictly lower-triangular in the ordered basis $(X^n, X^{n-1}Y, \dots, Y^n)$: the $m = 0$ row is the empty sum, identically zero, and the off-diagonal entries $\binom{n-j}{m-j}$ for $j < m$ are positive. The $m = 0$ row forces $[\delta_T P]_{X^n} = 0$ for every P , while rows $m = 1, \dots, n$ determine $p_0, \dots, p_{n-1}$ uniquely from any prescribed right-hand side, with p_n unconstrained. Hence $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$, and the solution P is unique modulo $\mathbb{C} \cdot Y^n$.

(c) The moment expansion (14) gives $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} I_T^0(f, \tau_0)$ where $I_T^0(f, \tau_0) = \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau$. Substituting the Fourier expansion $f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau}$, absolutely convergent on the compact T -segment, and integrating term by term,

$$\int_{\tau_0-1}^{\tau_0} e^{2\pi i n \tau} d\tau = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0, \end{cases}$$

by T -periodicity over a unit segment. Hence $I_T^0(f, \tau_0) = c_f(0)$ for every admissible τ_0 , which proves (17).

Combining (b) and (c) gives the solvability statement. $\square$

Proposition 3.2 (Bernoulli closed form for δ_T^{-1}). *For every $Q \in V_n$ with vanishing X^n -coefficient there exists $P \in V_n$ unique modulo $\mathbb{C} \cdot Y^n$ with $\delta_T P = Q$. Writing $Q = \sum_{r=1}^n q_r X^{n-r} Y^r$ and $P = \sum_{m=0}^n p_m X^{n-m} Y^m$, the solution is given in closed form by*

$$p_m = \sum_{r=1}^{m+1} \frac{B_{m-r+1} (n-r)!}{(m-r+1)! (n-m)!} q_r \quad (m = 0, 1, \dots, n-1), \quad (19)$$

where B_ℓ are Bernoulli numbers ($B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_{2\ell+1} = 0$ for $\ell \geq 1$); the coefficient p_n is unconstrained and parametrises $\ker(\delta_T) = \mathbb{C} \cdot Y^n$.

Proof. The operator identity $T = e^{Y\partial_X}$ on V_n (since T acts as the translation $X \mapsto X + Y$ by Y) gives $\delta_T = e^{Y\partial_X} - 1$. The Bernoulli generating function $z/(e^z - 1) = \sum_{k \geq 0} (B_k/k!) z^k$ supplies the closed-form inverse

$$\delta_T^{-1} = (Y\partial_X)^{-1} \frac{Y\partial_X}{e^{Y\partial_X} - 1} = \sum_{k \geq 0} \frac{B_k}{k!} (Y\partial_X)^{k-1}, \quad (20)$$

in which the $k = 0$ summand is the formal antiderivative $(Y\partial_X)^{-1}$, acting on monomials by $(Y\partial_X)^{-1}(X^a Y^b) = X^{a+1} Y^b / (a+1)$ (for $b \geq 1$), and the $k \geq 1$ summands are differential operators that eventually annihilate any polynomial of bounded X -degree. Applied to $X^{n-r} Y^r$ for $r \geq 1$,

$$\delta_T^{-1}(X^{n-r} Y^r) = \sum_{k=0}^{n-r+1} \frac{B_k}{k!} \frac{(n-r)!}{(n-r-k+1)!} X^{n-r-k+1} Y^{r+k-1}. \quad (21)$$

Summing $Q = \sum_r q_r X^{n-r} Y^r$ termwise and reading off the $X^{n-m} Y^m$ coefficient identifies $k = m - r + 1$ (so r runs over $\{1, \dots, m+1\}$, k over $\{0, \dots, m\}$), yielding (19). Existence and uniqueness modulo $\mathbb{C} \cdot Y^n$ are Lemma 3.1(a, b).

The closed form (19) is verified against the LU-decomposition recursion of (18) for $f \in \{\Delta, \hat{\Delta}\}$ in the companion script `verify_periods.py` [4] to relative residual $\sim 10^{-40}$ at 50-digit `mpmath` precision. $\square$

Proposition 3.3 (Universal kernels: τ^{j-1} and $\tilde{k}_T(\tau, j)$). *Let $f \in S_k^!$ and $\tau_0 \in \mathbb{H}$. Let $P \in V_n / \mathbb{C} Y^n$ be the unique solution of $\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)$ from Lemma 3.1 and Proposition 3.2, and define the modified-cocycle S -value*

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P) \in V_n \otimes \mathbb{C}. \quad (22)$$

For each interior $j \in \{2, \dots, k-2\}$, the coefficient of the monomial $X^{n-(j-1)} Y^{j-1}$ in r_f admits the integral representation

$$[r_f]_{X^{n-(j-1)} Y^{j-1}} = (2\pi i)^{n+1} (-1)^{j-1} \binom{n}{j-1} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{j-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, j) d\tau \right], \quad (23)$$

where $\tilde{k}_T(\tau, s)$ is the Hurwitz-zeta kernel of (3) and the integral kernels — bare Mellin τ^{j-1} on the S -segment and bare Hurwitz $\tilde{k}_T(\tau, j)$ on the T -segment — are precisely the integer- $s = j$ specialisations of the all- s kernels τ^{s-1} and $k_T(\tau, s)$ of the L -function functional (6). At integer $s = j \in \{2, \dots, k-2\}$, $\tilde{k}_T(\tau, j)$ collapses to the $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, j) = -\frac{B_j(\tau+1)}{j} + (-1)^{j-1} \frac{B_{k-j}(\tau+1)}{k-j} \in \mathbb{Q}[\tau], \quad (24)$$

of degree $\leq k-2$, via the Hurwitz collapse $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ for $m \geq 0$ together with $e^{i\pi(j-1)} = (-1)^{j-1}$.

Combined with the Eichler–Shimura / Brown cuspidal-basis decomposition of r_f ([3, 13]) — which sets $[r_f]_{X^{n-(j-1)} Y^{j-1}} = \omega^\pm(f) c_S^{j-1}$ with $c_S^{j-1} := [p_f^\pm]_{X^{n-(j-1)} Y^{j-1}}$ Brown’s basis coefficient at the matched monomial — this proves identity (4) of Theorem 1.1.

Proof. S-piece. The moment expansion (14) gives

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{j=0}^n \binom{n}{j} (-1)^j I_S^j(f, \tau_0) X^{n-j} Y^j,$$

so $[C_S^f(\tau_0)]_{X^{n-(j-1)} Y^{j-1}} = (2\pi i)^{n+1} \binom{n}{j-1} (-1)^{j-1} I_S^{j-1}(f, \tau_0)$, with $I_S^{j-1} = \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{j-1} d\tau$; this contributes the S -segment piece of (23) against the bare Mellin kernel τ^{j-1} .

T-piece, Bernoulli form. S acts on V_n as $(X, Y) \mapsto (-Y, X)$, so for $P = \sum_l p_l X^{n-l} Y^l$ one has $[P|_S - P]_{X^{n-(j-1)} Y^{j-1}} = (-1)^{j-1} p_{n-j+1} - p_{j-1}$ (n even). By Proposition 3.2 each p_l is an explicit $\mathbb{Q}$ -linear combination of the T -moments I_T^r , $r = 1, \dots, n$, packaging $C_T^f(\tau_0)$; substituting and collecting on the T -integration side yields a $\mathbb{Q}$ -polynomial $\tilde{k}_T(j, \tau) \in \mathbb{Q}[\tau]$ of degree $\leq n$ such that the T -segment piece of $[r_f]_{X^{n-(j-1)} Y^{j-1}}$ reads $(2\pi i)^{n+1} (-1)^{j-1} \binom{n}{j-1} \int_{\gamma_{\tau_0}^T} f(\tau) [\tilde{k}_T(j, \tau) / \binom{n}{j-1} (-1)^{j-1}] d\tau$.

T-piece, Hurwitz form. The shift-difference equation (27) characterises $\tilde{k}_T(\tau, s)$ as the unique-modulo- $\mathbb{C}$ -constant kernel whose T -integral against any $f \in S_k^!$ implements the coboundary modification of Step 2–Step 5; at integer $s = j$, the Hurwitz collapse $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$

produces (24). The Bernoulli closed-form $\widehat{k}_T(j, \tau)$ of the preceding paragraph and the Hurwitz collapse $\binom{n}{j-1}(-1)^{j-1}\widetilde{k}_T(\tau, j)$ are two representatives of the same coboundary-correction polynomial: their difference lies in the $\mathbb{C} \cdot Y^n$ Eisenstein direction of the kernel of δ_T (Lemma 3.1(a)), and contributes only to the boundary monomials X^n, Y^n of r_f , not to the interior monomial $X^{n-(j-1)}Y^{j-1}$ for $j-1 \in \{1, \dots, n-1\}$. Hence $\int_{\gamma_{\tau_0}^T} f(\tau) [\widehat{k}_T(j, \tau) / \binom{n}{j-1}(-1)^{j-1}] d\tau = \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, j) d\tau$, yielding (23) with the bare Hurwitz kernel $\widetilde{k}_T(\tau, j)$ inside.

Independence of f and τ_0 : every combinatorial coefficient on the right-hand side of (23)–(24) depends only on n, j , and Bernoulli numbers, not on f or τ_0 .

Numerical verification of (23) at $k = 12$, $f \in \{\Delta, \widehat{\Delta}\}$, $j = 2$ (parity $-$) reproduces Brown's $\omega^-(\Delta) = -5\,585\,015.379\,310\,401\,866\dots i$ to relative residual $\sim 10^{-42}$ at 50-digit `mpmath` precision (`verify_periods.py` [4]); all nine interior $j \in \{2, \dots, 10\}$ extractions are in lockstep to comparable precision. $\square$

Corollary 3.4 (Lockstep and basepoint independence). *The right side of (4) is independent of $\tau_0 \in \mathbb{H}$ and of the interior index $j \in \{2, \dots, k-2\}$ within its parity sector. At each k with $\dim S_k = 1$, this yields a $(k-3)$ -fold redundant extraction of $\omega^\pm(f)$ from the Fourier coefficients of f , reflecting the 1-dimensionality of the cuspidal direction in $H^1(\Gamma; V_{k-2})$.*

Proof. τ_0 -independence. By (22), r_f depends on τ_0 only through the cohomology class $[C^f] \in H^1(\Gamma; V_n \otimes \mathbb{C})$ and the choice of coboundary representative P . The class is τ_0 -invariant by (16) (valid for f without interior poles), and the coboundary representative $P(\tau_0)$ from Proposition 3.2 is the unique-modulo- $\mathbb{C}Y^n$ solution of $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$. The residual $\mathbb{C}Y^n$ ambiguity lies in the Eisenstein direction: $Y^n|_S - Y^n = X^n - Y^n$ is a coboundary that contributes only to the boundary monomials X^n, Y^n of r_f . Therefore the cuspidal coordinates of r_f — the $\omega^\pm(f)$ — are τ_0 -invariant.

j -independence. By the Eichler–Shimura decomposition and Brown's basis ([3, 13]), r_f projects onto the 1-dimensional cuspidal direction as $\omega^+(f)p_f^+ + \omega^-(f)p_f^-$ modulo Eisenstein contributions. Reading off the coefficient of any interior monomial $X^{n-(j-1)}Y^{j-1}$ for $j \in \{2, \dots, k-2\}$ returns $\omega^\pm(f)c_S^{j-1}$ with $c_S^{j-1} := [p_f^\pm]_{X^{n-(j-1)}Y^{j-1}}$ the basis-polynomial coefficient. The prefactor $(2\pi i)^{k-1}(-1)^{j-1}\binom{k-2}{j-1}/c_S^{j-1}$ in (4) is exactly the basis-conversion factor together with the moment-expansion combinatorial structure that normalises the right side to $\omega^\pm(f)$ independently of j .

At $k = 12$ this gives a 9-fold redundancy ($j \in \{2, \dots, 10\}$); at general k with $\dim S_k = 1$, the interior range $\{2, \dots, k-2\}$ has $k-3$ elements. $\square$

Proof of Theorem 1.1. The proof proceeds in eight steps. Steps 1–3 motivate the operator δ_T as the natural object whose inversion delivers the cuspidal cocycle representative. Step 4 (Proposition 3.2) performs the inversion in closed form. Steps 5–7 read $\omega^\pm(f)$ off the resulting period polynomial. Step 8 connects this integer- s construction to the all- s Hurwitz-zeta extension proved in Theorem 1.2.

Step 1: the canonical cuspidal representative has $C_T^f = 0$. Within the cohomology class $[C^f] \in H^1(\Gamma; V_n)$, the representative with vanishing T -component is structurally distinguished. When $C_T = 0$, the cocycle relations $C_{S^2} = 0$ and $C_{(ST)^3} = 0$ collapse onto C_S alone, and the surviving relations on C_S are exactly the two Eichler–Shimura period-polynomial relations $r_f|(1+S) = 0$ and $r_f|(1+U+U^2) = 0$ with $U = TS$. The $C_T = 0$ representative is therefore the one whose S -value lies in the Eichler–Shimura period-polynomial space, where Brown's basis $\{p_f^+, p_f^-, \text{Eisenstein}\}$ is defined and the cuspidal coordinates are $\omega^\pm(f)$. Killing C_T^f is the structural goal of the proof, not a computational convenience.

Step 2: kill C_T^f by a coboundary modification. The only freedom inside the cohomology class is to add a coboundary $\delta P(\gamma) := P|_\gamma - P$ for some $P \in V_n$. Under this modification, $C_T^f(\tau_0) \mapsto C_T^f(\tau_0) - (P|_T - P)$. Killing C_T^f becomes the equation

$$\delta_T P := P|_T - P = C_T^f(\tau_0), \quad (25)$$

to be solved for $P \in V_n$. This identifies the linear endomorphism $\delta_T: V_n \rightarrow V_n$ of Lemma 3.1 as the precise object whose inversion delivers the desired coboundary representative.

Step 3: solvability of $\delta_T P = C_T^f$ is exactly cuspidality of f . Lemma 3.1 computes $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$ and shows that $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$. Solvability of (25) is therefore equivalent to $c_f(0) = 0$, i.e. to f being a (weak) cusp form. The cuspidality hypothesis $f \in S_k^!$ is not a separate input to the construction: it is the solvability condition of the coboundary equation in Step 2.

Step 4: closed-form Bernoulli inversion of δ_T . The operator identity $T = e^{Y\partial_X}$ (since T acts on V_n as the translation $X \mapsto X + Y$ by Y) gives $\delta_T = e^{Y\partial_X} - 1$, and the Bernoulli generating function $z/(e^z - 1) = \sum_{k \geq 0} B_k z^k / k!$ supplies δ_T^{-1} as a finite operator series on V_n (terminating because ∂_X annihilates polynomials of bounded X -degree after finitely many applications). Proposition 3.2 records the explicit closed-form expression for $P \in V_n / \mathbb{C}Y^n$ in terms of Bernoulli numbers. The appearance of Bernoulli numbers is forced by the operator identity; it is not coincidence but inevitability.

Step 5: the modified cocycle and its S -component r_f . With $P := \delta_T^{-1}((2\pi i)^{-(n+1)} C_T^f(\tau_0)) \in V_n / \mathbb{C}Y^n$ from Step 4, the modified cocycle $C'^f := C^f - (2\pi i)^{n+1} \delta P$ has $C_T'^f = 0$ by construction, and its S -value (22) — which we denote $r_f := C_S'^f(\tau_0)$ — is the Eichler–Shimura period polynomial of f . Brown’s cuspidal-basis decomposition ([3, 13]) reads

$$r_f = \omega^+(f) p_f^+ + \omega^-(f) p_f^- + (\text{Eisenstein-direction term}), \quad (26)$$

identifying $\omega^\pm(f)$ as explicit linear coordinates of r_f in the 1-dimensional cuspidal direction.

Step 6: unpacking r_f into bare Mellin and Hurwitz integrals. Substituting the moment expansion of $C_S^f(\tau_0)$ from (14) and the Bernoulli expression for P from Step 4 into (22), and reading off the coefficient of $X^{n-(j-1)} Y^{j-1}$ for each interior $j \in \{2, \dots, k-2\}$, Proposition 3.3 produces (4) with the bare Mellin kernel τ^{j-1} on the S -segment and the bare Hurwitz kernel $\tilde{k}_T(\tau, j)$ on the T -segment — the integer- $s = j$ specialisations of the L -function kernels of Theorem 1.2. All combinatorial structure $((2\pi i)^{k-1}, (-1)^{j-1}, \binom{k-2}{j-1})$ and the Brown basis-coefficient c_S^{j-1} sit in the prefactor outside the bracket.

Step 7: lockstep and basepoint independence. Both follow from r_f being a single fixed polynomial in V_n that lies in the 1-dimensional cuspidal direction. Reading off any interior monomial coefficient of r_f returns a constant multiple of the same $\omega^\pm(f)$ (parity-selected by j), and the prefactor $(-1)^{j-1} \binom{k-2}{j-1} (2\pi i)^{k-1} / c_S^{j-1}$ in (4) normalises away the basis-conversion factor. Independence in τ_0 follows from $[C^f]$ being a cohomology-class invariant; the residual $\mathbb{C}Y^n$ ambiguity in the coboundary representative P contributes only to the Eisenstein direction. Corollary 3.4 makes both statements precise; at $k = 12$, the 9 interior values of j give the 9-fold redundant extraction.

Step 8: bridge to Theorem 1.2. The bare Mellin kernel τ^{j-1} on the S -segment is the integer- $s = j$ specialisation of τ^{s-1} , and the bare Hurwitz kernel $\tilde{k}_T(\tau, j)$ on the T -segment is the integer- $s = j$ specialisation of the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$ of (3), via the Hurwitz collapse $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ for $m \geq 0$. Theorem 1.1 is therefore the integer- s avatar of Theorem 1.2; the all- s extension is proved in §4. $\square$

4 The Hurwitz-zeta extension and the L -function on $M_k^!$: proof of Theorem 1.2

4.1 Derivation of the kernel $\tilde{k}_T$

We seek a kernel $\tilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that $L^*(f, s; \tau_0)$ as defined in (6) is independent of τ_0 for every $f \in M_k^!$. Differentiating (6) in τ_0 and using $\Delta(-1/\tau_0) = \tau_0^k \Delta(\tau_0)$ (the S -modularity of f at the S -segment endpoint $-1/\tau_0$) together with $f(\tau_0 - 1) = f(\tau_0)$ (the T -modularity at $\tau_0 - 1$) gives, after cancelling the common factor $i^{-s} f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}. \quad (27)$$

The right side picks up the S -segment endpoint contribution at $-1/\tau_0 = e^{i\pi} \tau_0^{-1}/(-1) = e^{i\pi}/\tau_0$ via the branch identity $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ combined with the Jacobian $d(-1/\tau_0)/d\tau_0 = \tau_0^{-2}$ and the modularity factor τ_0^k , giving $e^{i\pi(s-1)} \tau_0^{k-1-s}$ after simplification.

Equation (27) is the complex- s analog of the polynomial coboundary equation $\delta_T P = Q$ of §3. The Hurwitz raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma}$$

inverts the shift operator $\tau \mapsto \tau-1$ for any complex σ , on the analytic function $\zeta(\sigma, \tau+1)$. Applying at $(\sigma, a) = (1-s, \tau_0)$ and $(\sigma, a) = (s-(k-1), \tau_0)$, and combining with the sign $e^{i\pi(s-1)}$ gives (3).

Lemma 4.1. *For $f \in M_k^!$, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ defined by (6) is independent of $\tau_0 \in \mathbb{H}$ for every $s \in \mathbb{C}$.*

Proof. Direct τ_0 -differentiation, using (27) to cancel the boundary contributions from the two segments. Both integrals in (6) are finite at every $\tau_0 \in \mathbb{H}$ because f has no poles in $\mathbb{H}$ and the kernels τ^{s-1} , $\tilde{k}_T(\tau, s)$ are analytic in $\tau \in \mathbb{H}$ for every fixed $s \in \mathbb{C}$ (the Hurwitz $\zeta(\sigma, a)$ is analytic in a on $\text{Re}(a) > 0$). $\square$

4.2 The Hurwitz-unfolding lemma

Lemma 4.2 (Hurwitz unfolding). *For every positive integer n and every $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (28)$$

Proof. First assume $\text{Re}(\nu) > 1$, where the Hurwitz series $\zeta(\nu, \sigma) = \sum_{m \geq 0} (\sigma + m)^{-\nu}$ converges uniformly on the compact segment $[i, i+1]$ [6, §25.11]. Series-integral exchange and the substitution $u = \sigma + m$ in the m -th summand collapse the sum to a single integral by integer- n periodicity $e^{2\pi i n m} = 1$:

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = \sum_{m \geq 0} \int_{i+m}^{i+m+1} e^{2\pi i n u} u^{-\nu} du = \int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du,$$

an integral along the horizontal ray $\{i+t : t \geq 0\} \subset \mathbb{H}$.

Substitute $t = -2\pi i n u$: $u = it/(2\pi n)$, $du = i dt/(2\pi n)$, and $u^{-\nu} = i^{-\nu}(2\pi n)^{\nu} t^{-\nu}$ on the principal branch (consistent with $\arg(u) = \pi/2$ at $u = i$ and $\arg(t) = 0$ at $t = 2\pi n$). The contour in t runs from $2\pi n$ down to $2\pi n - i\infty$, in the right half-plane. Since $e^{-t} t^{-\nu}$ decays exponentially as $|t| \rightarrow \infty$ in $\text{Re}(t) > 0$, Cauchy's theorem applied to the rectangle $\{2\pi n \leq \text{Re}(t) \leq R, -R \leq \text{Im}(t) \leq 0\}$ in the limit $R \rightarrow \infty$ deforms the vertical contour $[2\pi n, 2\pi n - i\infty]$ to the real-axis contour $[2\pi n, +\infty]$:

$$\int_{2\pi n}^{2\pi n - i\infty} e^{-t} t^{-\nu} dt = \int_{2\pi n}^{\infty} e^{-t} t^{-\nu} dt = \Gamma(1 - \nu, 2\pi n)$$

by the integral definition of the upper incomplete gamma [6, §8.2]. Collecting prefactors gives (28) for $\text{Re}(\nu) > 1$.

Both sides of (28) are entire functions of ν : the upper incomplete gamma $\Gamma(s, x)$ is entire in s for fixed $x \neq 0$ [6, §8.2.7], and the integral on the left is entire in ν because $\zeta(\sigma, \sigma_0)$ is meromorphic in σ with a simple pole only at $\sigma = 1$, whose residue contribution to $\int_{i-1}^i e^{2\pi i n \tau} d\tau = 0$ for nonzero integer n (since $e^{2\pi i n(i-1)} = e^{2\pi i n i}$). Analytic continuation in ν extends (28) to all complex ν . $\square$

4.3 Per-mode evaluation: proof of Theorem 1.2

By Lemma 4.1, the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L^*(f, s)$ is common to all admissible contours and basepoints. At the elliptic fixed-point basepoint $\tau_0 = i$, the S -segment $[-1/i, i] = [i, i]$ has zero length and the S -integral in (6) vanishes identically, leaving

$$L^*(f, s; \gamma_{\tau_0}^S(i) = 0, \gamma_{\tau_0}^T(i)) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau.$$

Expanding $f = \sum_n c_f(n) q^n$ as a q -series and swapping sum and integral (justified by uniform convergence of the q -series on the compact segment $[i-1, i]$, including any finite number of polar modes since $|q^{-1}| = e^{2\pi}$ is bounded there) gives

$$L^*(f, s) = i^{-s} \sum_n c_f(n) \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau. \quad (29)$$

We treat the constant Fourier mode $n = 0$ and the non-constant modes $n \neq 0$ separately.

The constant Fourier mode. For $n = 0$, the integrand reduces to $\tilde{k}_T(\tau, s)$ itself. The Hurwitz zeta has antiderivative $\int \zeta(\sigma, a) da = -\zeta(\sigma - 1, a)/(\sigma - 1) + C$ for $\sigma \neq 1$, and the Hurwitz raising identity $\zeta(\sigma - 1, a + 1) - \zeta(\sigma - 1, a) = -a^{1-\sigma}$ at $a = i$ gives

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma - 1}. \quad (30)$$

Applying (30) to each half of $\tilde{k}_T$ at $\sigma = 1 - s$ and $\sigma = s - (k - 1)$:

$$\int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -\frac{i^s}{s} + e^{i\pi(s-1)} \frac{i^{k-s}}{k-s}.$$

The overall i^{-s} prefactor and the phase collapse $i^{-s} e^{i\pi(s-1)} i^{k-s} = -i^k$ give the constant-term contribution

$$c_f(0) \cdot i^{-s} \int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right). \quad (31)$$

This matches the polar term of [2, Lemma 2.1] (equivalently, the polar piece of [1, Thm. 1.1]) at the splitting point $B = 1$.

The non-constant Fourier modes. For $n \neq 0$, substituting $\sigma = \tau + 1$ shifts the contour to $[i, i + 1]$ with the prefactor $e^{-2\pi i n} = 1$. Apply Lemma 4.2 to each half of $\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)}\zeta(s - (k - 1), \tau + 1)$, i.e. at $\nu = 1 - s$ and $\nu = s - (k - 1)$:

$$\begin{aligned} \int_i^{i+1} e^{2\pi i n \sigma} \zeta(1 - s, \sigma) d\sigma &= i^s (2\pi n)^{-s} \Gamma(s, 2\pi n), \\ \int_i^{i+1} e^{2\pi i n \sigma} \zeta(s - (k - 1), \sigma) d\sigma &= i^{k-s} (2\pi n)^{s-k} \Gamma(k - s, 2\pi n). \end{aligned}$$

The first half against the i^{-s} prefactor gives $\Gamma(s, 2\pi n)/(2\pi n)^s$; the second half against the relative phase $-e^{i\pi(s-1)}$ collapses via $-i^{-s} e^{i\pi(s-1)} i^{k-s} = i^k$ to $i^k \Gamma(k - s, 2\pi n)/(2\pi n)^{k-s}$. Adding the two halves gives the per-mode summand

$$c_f(n) \cdot i^{-s} \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau = c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k - s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (32)$$

which is the $B = 1$ value of the cuspidal summand of [2, Lemma 2.1].

Summing (31) and (32) over all Fourier modes reproduces the closed-form sum (7).

4.4 Worked example at $k = 12$: Bernoulli polynomials and numerics

At $k = 12$ the kernel of (3) is

$$\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - 11, \tau + 1).$$

At integer values of $s \in \{1, 2, \dots, 11\}$, both Hurwitz zeta factors collapse to Bernoulli polynomials via $\zeta(-m, a) = -B_{m+1}(a)/(m + 1)$ ($m \geq 0$), and $\tilde{k}_T(\tau, s)$ becomes a polynomial of degree ≤ 11 in τ . Three representative values:

$$\begin{aligned} \tilde{k}_T(\tau, 1) &= -\tau - \frac{1}{2} + \frac{1}{11} B_{11}(\tau + 1), & \tilde{k}_T(\tau, 6) &= -\frac{1}{3} B_6(\tau + 1), \\ \tilde{k}_T(\tau, 11) &= -\frac{1}{11} B_{11}(\tau + 1) + \tau + \frac{1}{2}. \end{aligned}$$

The $s \leftrightarrow 12 - s$ symmetry follows from the $s \leftrightarrow k - s$ structure of the two-term kernel (with $i^k = +1$ at $k = 12$).

Identification with Theorem 1.1. With the clean form of Theorem 1.1, the T -segment kernel of (4) is exactly the integer- $s = j$ specialisation $\tilde{k}_T(\tau, j)$ of the Hurwitz-zeta kernel of (3), with no convention conversion required: all combinatorial structure $(-1)^{j-1} \binom{k-2}{j-1} / c_S^{j-1}$ governing the $\omega^\pm(f)$ normalisation lives outside the two-segment bracket. The period-extraction functional (4) is thus the rescaled integer- s shadow of the all- s L -function functional (6) of Theorem 1.2, with the bare Mellin τ^{s-1} and Hurwitz $\tilde{k}_T(\tau, s)$ kernels mapped to their integer- $s = j$ specialisations and the analytic prefactor i^{-s} replaced by the combinatorial Brown-basis factor.

Numerical values. Running the companion `mpmath` scripts [4] at 50-digit precision, with $\tau_0 \in \{0.3 + 1.2i, i, 1.5i, -0.4 + 0.8i\}$ and $f \in \{\Delta, \hat{\Delta}\}$, reproduces the values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ of [3]:

$$\begin{aligned} \omega^+(\Delta) &\approx -68916772.8096\dots, & \omega^-(\Delta)/i &\approx -5585015.3793\dots, \\ \eta^+(\hat{\Delta}) &\approx 1.27202 \times 10^{11}, & \eta^-(\hat{\Delta})/i &\approx 1.02767 \times 10^{10}, \end{aligned}$$

with relative residuals $\lesssim 10^{-40}$, exhibiting lockstep across $j \in \{2, \dots, 10\}$ and basepoint-independence across all four τ_0 choices.

5 Wall-crossing for meromorphic forms: proof of Theorem 1.3

5.1 Existence and local constancy

Let f be a meromorphic modular form of even weight k on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint such that the contour pair (1) avoids the Γ -orbit of $f^{-1}(\infty) \cap \mathbb{H}$. The integrands $f(\tau) \tau^{s-1}$ and $f(\tau) \tilde{k}_T(\tau, s)$ are continuous on each compact segment for every fixed $s \in \mathbb{C}$ (the kernels are analytic in $\tau \in \mathbb{H}$), so both integrals in (2) are finite and $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is well-defined. This proves Theorem 1.3(a).

For local constancy, the argument of §4.1 applies verbatim: differentiating (2) in τ_0 gives the boundary contributions

$$i^{-s} f(\tau_0) \left[\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) + \tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s} \right] = 0$$

by (27), provided the modularity identities $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ and $f(\tau_0 - 1) = f(\tau_0)$ hold at the contour endpoints (which they do for every meromorphic modular form, irrespective of the location of interior poles). Hence $dL^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)/d\tau_0 = 0$ whenever the contour pair varies within the admissible open subset of $\mathbb{H}$. This establishes Theorem 1.3(b).

5.2 Wall-crossing across an interior pole

Suppose a continuous one-parameter deformation of τ_0 to τ'_0 in $\mathbb{H}$ moves the S -segment $[-1/\tau_0, \tau_0]$ across an interior pole $\tau_p \in \mathbb{H}$ of f (the T -segment case is identical up to the choice of kernel; see below). Concretely, let $\Sigma_t \subset \mathbb{H}$ denote the S -segment at parameter $t \in [0, 1]$, with Σ_0 the S -segment at τ_0 and Σ_1 at τ'_0 , and assume that the swept-out region $\bigcup_t \Sigma_t$ encloses τ_p with winding number $\sigma_p^S \in \{+1, -1\}$ (orientation of the crossing). By Cauchy's theorem applied to the closed contour $\Sigma_1 - \Sigma_0$,

$$\int_{\Sigma_1} f(\tau) \tau^{s-1} d\tau - \int_{\Sigma_0} f(\tau) \tau^{s-1} d\tau = 2\pi i \sigma_p^S \operatorname{Res}(f, \tau_p) \tau_p^{s-1}$$

for a simple pole of f at τ_p . An identical argument applies when the T -segment $[\tau_0 - 1, \tau_0]$ sweeps across τ_p , yielding

$$\int_{T\text{-seg at } \tau'_0} - \int_{T\text{-seg at } \tau_0} = 2\pi i \sigma_p^T \operatorname{Res}(f, \tau_p) \tilde{k}_T(\tau_p, s),$$

where $\sigma_p^T \in \{+1, -1\}$ encodes the orientation of the T -segment crossing. Combining (with the i^{-s} prefactor) gives

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_p^S \tau_p^{s-1} + \sigma_p^T \tilde{k}_T(\tau_p, s) \right]. \quad (33)$$

This is the residue formula $\mathcal{R}_{\tau_p}(f, s)$ of (10), with the winding-number indices σ_p^S, σ_p^T making explicit the dependence on which segment(s) cross. At a generic wall only one of σ_p^S, σ_p^T is nonzero; both can be nonzero on the codimension-two loci where the S - and T -segments simultaneously cross Γ -images of τ_p .

For a pole of order $m \geq 1$ at τ_p with Laurent expansion $f(\tau) = \sum_{\ell \geq -m} r_{f, \tau_p}(\ell) (\tau - \tau_p)^\ell$, the residue formula (33) generalises by replacing $\operatorname{Res}(f, \tau_p) K(\tau_p, s)$ with $\sum_{\ell=1}^m r_{f, \tau_p}(-\ell) \partial_{\tau_p}^{\ell-1} K(\tau_p, s) / (\ell-1)!$ for each relevant kernel $K \in \{\tau^{s-1}, \tilde{k}_T(\tau, s)\}$, by the standard residue formula for higher-order poles.

5.3 Reduction on regular-at-cusps meromorphic forms

Let $f \in F_k$ be regular at the cusp $i\infty$, so $f(\tau) = \sum_{n \geq N} c_f(n) q^n$ for some $N \geq 0$.

Lemma 5.1 (Cusp decay of the T -segment). *For $f \in F_k$ regular at the cusp with $f(\tau) = O(q^N)$ at $i\infty$ for $N \geq 1$, every $s \in \mathbb{C}$, and every $\varepsilon \in \mathbb{R}$ fixed,*

$$\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = O(|\tau_0|^{C(\operatorname{Re} s)} e^{-2\pi N \operatorname{Im}(\tau_0)}) \quad \text{as } \operatorname{Im}(\tau_0) \rightarrow \infty, \operatorname{Re}(\tau_0) = \varepsilon.$$

The exponent $C(\operatorname{Re} s) = \max(0, 1/2 - \operatorname{Re}(1-s)) + \max(0, 1/2 - \operatorname{Re}(s - (k-1))) + \varepsilon_0$ for any $\varepsilon_0 > 0$ is controlled by the standard polynomial-growth bound for the Hurwitz zeta on vertical strips [6, §25.11.36]. When $N = 0$ (i.e. f has a nonzero constant term at $i\infty$), the T -segment is $O(|\tau_0|^{C(\operatorname{Re} s)})$ but does not decay; in this case the constant-term Fourier mode must be split off and treated separately (see [2, Lemma 2.1] for the analogous polynomial correction in the deformed-Mellin construction).

Proof. On the segment $\tau \in [\tau_0 - 1, \tau_0]$ with $\operatorname{Im}(\tau) = \operatorname{Im}(\tau_0)$, the bound $|f(\tau)| \leq M e^{-2\pi N \operatorname{Im}(\tau_0)}$ holds for some $M = M(f)$ depending only on f . Combining with the polynomial-in- τ growth of $\tilde{k}_T(\tau, s)$ from [6, §25.11.36] and integrating over the length-1 segment gives the stated bound. $\square$

To match the deformed-Mellin construction of $L_f^*(s)_{\text{int}}$ (see [2]), take $\tau_0 = \varepsilon + i\Lambda$ for some small $\varepsilon \in \mathbb{R}$ chosen so that $\operatorname{Re}(\tau_0) = \varepsilon$ avoids all Γ -images of interior poles of f . As $\Lambda \rightarrow \infty$:

- The T -segment integral $\rightarrow 0$ by Lemma 5.1 (using $N \geq 1$; for $N = 0$ the analog argument with the constant-term subtraction $c_f(0)$ from [2] applies).
- The S -segment $[-1/\tau_0, \tau_0]$ approaches the $\Lambda \rightarrow \infty$ limit of the Λ -truncated Mellin contour along $\operatorname{Re}(\tau) = \varepsilon$ (modulo orientation, which fixes the L_{int}^* convention).

By Lemma 4.1 (extended to meromorphic f within a fixed connected component, by §5.1 above), the left side is Λ -independent, so

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \lim_{\Lambda \rightarrow \infty} L^*(f, s; \gamma_{\varepsilon+i\Lambda}^S, \gamma_{\varepsilon+i\Lambda}^T) = L_f^*(s)_{\text{int}, \gamma_\varepsilon},$$

where γ_ε is the deformed-Mellin contour along $\operatorname{Re}(\tau) = \varepsilon$. This proves Corollary 1.4. $\square$

5.4 Wall-crossing of the period-polynomial cocycle

The wall-crossing residue formula (10) acts on the complex- s functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$. Specialised at integer $s = j \in \{2, \dots, k-2\}$ and assembled into the V_{k-2} moment expansion of §2, it gives the wall-crossing of the period-polynomial cocycle of f under contour deformation across an interior pole.

Corollary 5.2 (Period-polynomial wall-crossing residue). *Let f be a meromorphic modular form of weight k on Γ , and let $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T), (\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ be two admissible contour pairs related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f . Denote by $C_S^f(\gamma_{\tau_0}^S), C_T^f(\gamma_{\tau_0}^T) \in V_{k-2} \otimes \mathbb{C}$ the corresponding V_{k-2} -valued cocycle values from (12). For a simple pole $\operatorname{Res}(f, \tau_p)$,*

$$C_S^f(\gamma_{\tau_0}^S) - C_S^f(\gamma_{\tau'_0}^S) = 2\pi i (2\pi i)^{k-2} \sigma_{\tau_p}^S \operatorname{Res}(f, \tau_p) (X - \tau_p Y)^{k-2}, \quad (34)$$

and analogously for C_T^f with $\sigma_{\tau_p}^S$ replaced by $\sigma_{\tau_p}^T$. The integer- $s = j$ specialisation of (10) is the coefficient of $\binom{k-2}{j} (-1)^j X^{k-2-j} Y^j$ in (34) (combined for S and T). For $\dim S_k = 1$ and $f \in F_k$ meromorphic with interior poles, the period polynomial r_f extracted via Theorem 1.1 from $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ therefore depends on the homotopy class of the contour pair, with jump (34) across each wall.

Proof. The integrand $f(\tau)(X-\tau Y)^{k-2}$ of (12) has a simple pole at $\tau = \tau_p$ with residue $\text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$, so a contour deformation crossing τ_p with winding $\sigma_{\tau_p} \in \{-1, +1\}$ contributes $2\pi i \sigma_{\tau_p} \text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$ to the moment integral. Multiplying by the cocycle prefactor $(2\pi i)^{k-2}$ of (12) gives (34). Expanding $(X - \tau_p Y)^{k-2}$ in the monomial basis and matching coefficients of $X^{k-2-j}Y^j$ for each $j \in \{0, \dots, k-2\}$ reproduces (10) at integer $s = j$ (with τ_p^{j-1} arising from $X^{k-2-j}Y^j$ at $j \geq 1$ from the S -segment contribution, and analogously for the T -segment via $\tilde{k}_T(\tau_p, j)$ at integer j). $\square$

5.5 Remark on the polylog-projection machinery

The polylog projection $\hat{P}_{\tau_p}: f \mapsto \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p))$ (see [2, §2.2]) subtracts the leading polar tails of f at each interior pole τ_p , ensuring that the Λ -truncated Mellin integral against f has a finite $\Lambda \rightarrow \infty$ limit (necessary in the deformed-Mellin construction because of the Hagedorn-rate Fourier coefficients $|c_f(n)| \sim n^{p-1} e^{2\pi n \text{Im}(\tau_p)}$ generated by interior poles [2, Prop. 1.1, 2.7]). In the finite-contour formulation (2) the $\Lambda \rightarrow \infty$ limit is not taken, so $\hat{P}_{\tau_p}$ is not required for the existence or local-constancy properties of L^* . The polylog-projection machinery does remain relevant, however, for the *closed-form per-mode evaluation* of L^* in incomplete gammas parallel to (7), where the Hagedorn-rate Fourier modes must be split off and dealt with separately before each mode contributes a clean incomplete-gamma term. The systematic interior-pole closed form parallel to Theorem 1.2 is left to the sequel.

6 Outlook

Wilson-loop / holonomy interpretation. The period-defining contour pair, projected to the modular curve $Y(1)$, descends to the two generators of the orbifold fundamental group $\pi_1^{\text{orb}}(Y(1)) \cong \mathbb{Z}/2 * \mathbb{Z}/3$. The polynomial coefficient system V_{k-2} is a flat local system over $Y(1)$, and the cocycle C_γ^f is the weighted holonomy of this flat connection along the loop γ , with f supplying the gauge-invariant weight. The cohomology class $[C^f] \in H^1(\Gamma; V_{k-2})$ is the analog of a gauge-invariant observable; τ_0 is the analog of a trivialisation frame; $\omega^\pm(f)$ are coordinates of $[C^f]$ in the cuspidal subspace, the analog of physical observables read off a Wilson-loop expectation. The period-defining finite-contour functional is, in this dictionary, the modular-form analog of a Wilson-loop holonomy on the modular curve.

The on-shell methods analogy. The classical Mellin construction is “off-shell” in the same sense as a Lagrangian gauge-theory amplitude calculation: the intermediate integrand $f(\tau) \tau^{s-1} d\tau$ encodes gauge-redundant information (the basepoint τ_0 , the contour to the cusp), and the gauge-invariant observable $L^*(f, s)$ is reached only after a non-canonical limit. Theorem 1.2 is an “on-shell” reformulation in which the gauge-invariant observable is the value of a finite-contour functional at any admissible basepoint, with no limit and no regulator. The modular-form analog of the on-shell S-matrix programme [15, 16] — constructing $\omega^\pm$ and $L^*(f, s)$ from $\{c_f(n)\}$ via a kernel that does not pass through the cocycle / coboundary scaffolding — is partially answered by Theorem 1.1 on the period side and Theorem 1.2 on the L -function side, with the remaining question being whether the kernels themselves admit a construction independent of the δ_T^{-1} Bernoulli operator.

Regularisation as a symptom of bad framing. The disappearance of any regularisation step in (2) is an instance of a recurring methodological pattern: *when a divergence arises only after one has made a non-canonical upstream choice, finding the canonical formulation eliminates the*

divergence together with the regulator. The same move underlies the on-shell amplitudes programme [15]. The pattern is not universal — chiral anomalies in even dimensions are a counterexample, with cohomologically robust “regulator” content invariant under reframings — but it is a useful prior, and the finite-contour formulation here is a particularly clean instance in which the canonical $L^*(f, s)$ falls out as the value of a well-defined linear functional at any admissible basepoint, with no auxiliary parameter to take to infinity and no contour to deform to a cusp.

The geometric content of the Mellin kernel. The polynomial coefficient system V_{k-2} underlying the period construction is the symmetric-power local system that gives the polynomial kernel $(X - \tau Y)^{k-2}$ of the period polynomial. Its flat connection on the modular curve is the natural geometric home of $\omega^\pm(f)$. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3) extends the polynomial kernel to a one-parameter analytic family in s , but the geometric content of the bare Mellin kernel $f(\tau) \tau^{s-1} d\tau$ itself — as opposed to its specialisation at integer s to a polynomial-in- V_{k-2} kernel — is not visible in the construction of this paper. A natural conjecture is that $\tau^{s-1} d\tau$ is a section of an infinite-dimensional principal-series-type local system on $Y(1)$, in the same one-parameter family as the Lewis–Zagier period functions for Maass wave forms [14]. Identifying the relevant local system explicitly, and connecting the Hurwitz-zeta inversion of δ_T to a coboundary statement in this larger system, is an open question.

Poincaré-series structure of the cumulative wall-crossing. The residue formula (10) controls the jump of $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ across a single pole-crossing. Summing such jumps along a continuous deformation of the contour pair from one that encloses no interior poles to one that encloses every pole in a single Γ -orbit $\Gamma \cdot \tau_p$ produces a cumulative residue of the structure

$$\sum_{\gamma \in \Gamma_{\tau_p} \backslash \Gamma} (c\tau_p + d)^{k-2} \text{Res}(f, \tau_p) \left[\sigma_\gamma^S (\gamma\tau_p)^{s-1} + \sigma_\gamma^T \tilde{k}_T(\gamma\tau_p, s) \right],$$

where the cocycle factor $(c\tau_p + d)^{k-2}$ comes from the transformation $\text{Res}(f, \gamma\tau_p) = (c\tau_p + d)^{k-2} \text{Res}(f, \tau_p)$ under modularity of f , Γ_{τ_p} is the stabiliser of τ_p in Γ , and $\sigma_\gamma^{S,T}$ encode the winding numbers of $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ across the orbit-point $\gamma\tau_p$. The summand is structurally a weight- k Poincaré-series term in τ_p , in the s -extended form built from τ_p^{s-1} and $\tilde{k}_T(\tau_p, s)$.

The structural parallel with the partial-fraction representation proved by Hardy and Ramanujan in their 1919 paper on modular functions with prescribed poles [12] is direct. [12, Thm. 2] shows that a modular function ϕ of weight n with a single simple pole at $\tau = \alpha$ admits the representation

$$f(q) = 2\pi i A \sum \frac{1}{(a + b\alpha)^{n+2}} \frac{\mathbf{q}^2}{q^2 - \mathbf{q}^2}, \quad \mathbf{q} = \exp\left(\frac{c+d\alpha}{a+b\alpha} \pi i\right),$$

summed over pairs (a, b) giving distinct Γ -images of α . Each summand is the residue at one orbit-point, transported by the modular cocycle factor $(a + b\alpha)^{-(n+2)}$ and weighted by an explicit kernel evaluated at $\mathbf{q}$. Our cumulative wall-crossing residue has the same orbit-of-residues structure, with $\tilde{k}_T(\gamma\tau_p, s)$ playing the role of the $\mathbf{q}^2/(q^2 - \mathbf{q}^2)$ kernel. Whether the cumulative wall-crossing converges as a function of s to a canonical weight- k Poincaré series, and what its modular properties are, is a natural direction for follow-up work.

Towards systematic interior-pole L -function theory. Theorem 1.3 establishes existence and wall-crossing for $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ on arbitrary meromorphic f , but does not give a closed-form per-mode evaluation parallel to the incomplete-gamma sum (7). Developing the systematic per-mode

theory — in particular adapting the polylog-projection machinery $\widehat{P}_{\tau_p}$ (see [2]) to the finite-contour setting and matching the wall-crossing data (9) to the iterated-Eichler-integral framework of Brown–Hain on $\mathcal{M}_{1,1}$ and the universal elliptic curve [3] — is the natural sequel. The arithmetic of the wall-crossing residues $\mathcal{R}_{\tau_p}(f, s)$ for forms with poles at CM points (e.g. $\tau_p \in \{i, \rho\}$) is a particularly clean target, where the Laurent coefficients of f take values in number fields and $\mathcal{R}_{\tau_p}$ acquires explicit arithmetic structure.

A Algebraic derivation of the Hurwitz-zeta kernel

The body of §4.1 derives $\widetilde{k}_T(\tau, s)$ by differentiating the two-segment functional in τ_0 and demanding the result vanish. Here we record the parallel derivation that proceeds *from* the classical Mellin transform of a holomorphic cusp form, splitting it at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding the resulting tail be representable as a T -segment integral. Both derivations produce the same shift-difference equation and the same kernel; this appendix makes the algebraic provenance of the Hurwitz-zeta combination manifest.

Step 1: Mellin split at the contour endpoints. Let $f \in S_k$ with k even. The classical completed L -function $L^*(f, s) = (2\pi)^{-s} \Gamma(s) L(f, s)$ has integral representation

$$L^*(f, s) = -i \int_0^{i\infty} f(\tau) (\tau/i)^{s-1} d\tau = i^{-s} \int_0^{i\infty} f(\tau) \tau^{s-1} d\tau,$$

the second equality using $(\tau/i)^{s-1} = i^{1-s} \tau^{s-1}$ on the imaginary axis. Pick $\tau_0 \in \mathbb{H}$ and split the imaginary-axis contour at the two endpoints $\{-1/\tau_0, \tau_0\}$ in (1). Reorganising the three resulting pieces using S -modularity $f(-1/\tau) = \tau^k f(\tau)$ on the lower segment $[0, -1/\tau_0]$ (via the substitution $\tau \mapsto -1/\tau$) and computing the q -series-against- τ^{s-1} integral on $[\tau_0, i\infty]$ directly yields, for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,

$$L^*(f, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (35)$$

The left side is τ_0 -independent (it is the classical L -value); both terms on the right side individually depend on τ_0 , but their sum does not. At $\tau_0 = i$ the argument $-2\pi i n \tau_0 = 2\pi n$ is real and (35) reduces to the standard BFK incomplete-gamma sum of [1].

The right side of (35) contains the S -segment integral $i^{-s} \int_{-1/\tau_0}^{\tau_0}$ explicitly; the remaining incomplete-gamma sum plays the role that we want to be a T -segment integral.

Step 2: ansatz for the T -segment kernel. Postulate that there exists a kernel $\widetilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \widetilde{k}_T(\tau, s) d\tau = \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right] \quad (36)$$

for every $f \in S_k$ and every $\tau_0 \in \mathbb{H}$. If such a kernel exists, substituting (36) into (35) gives the two-segment functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = i^{-s} [\int_{\gamma_{\tau_0}^S} + \int_{\gamma_{\tau_0}^T}]$ as a τ_0 -independent expression equal to $L^*(f, s)$.

Step 3: τ_0 -derivative gives a shift-difference equation. Differentiate both sides of (36) in τ_0 . On the left, the only τ_0 -dependence is in the integration limits, so $d/d\tau_0$ acts as

$$\frac{d}{d\tau_0} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = f(\tau_0) [\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s)],$$

using $f(\tau_0 - 1) = f(\tau_0)$ for the lower limit. On the right of (36), differentiating each incomplete gamma in its second argument (the explicit τ_0 -dependence) and assembling the Fourier series yields, after canceling a common factor of $f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s} \quad \text{for all } \tau_0 \in \mathbb{H}. \quad (37)$$

The first term on the right is the contribution of the τ^{s-1} -Mellin endpoint at τ_0 ; the second term is the S -modularity image of the same endpoint at $-1/\tau_0$, carrying the branch factor $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ together with the modularity Jacobian τ_0^k . Equation (37) is a shift-difference equation of order one in τ , with a sum of two power-of- τ terms on the right side.

Step 4: Hurwitz raising solves the shift-difference equation. The Hurwitz zeta function $\zeta(\sigma, a)$ satisfies the raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma} \quad (\sigma \in \mathbb{C}, \operatorname{Re}(a) > 0) \quad (38)$$

(see [6, §25.11.3]). Applied at $(\sigma, a) = (1-s, \tau)$ and $(\sigma, a) = (s-(k-1), \tau)$ respectively, this gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-(k-1), \tau+1) - \zeta(s-(k-1), \tau) = -\tau^{k-1-s}.$$

Combining with the relative phase $e^{i\pi(s-1)}$ from the second term of (37) gives the kernel

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1), \quad (39)$$

which by construction satisfies (37) and hence (36). The kernel is analytic in τ on $\mathbb{H}$ (both Hurwitz zetas have second argument $\tau+1$ with $\operatorname{Re}(\tau+1) > 0$) and meromorphic in s with simple poles only at $s=0$ and $s=k$, where the per-mode integrals collect removable cancellations (cf. Lemma 4.2). Equation (39) is the kernel (3) of the main text.

Where DR2022 stops. The kernel (39) has two Hurwitz-zeta terms. The construction of [8, Thm. 1.2] gives the single-term

$$\tilde{k}_T^{\text{DR}}(\tau, m) = \zeta^*(1-m, \tau+1)$$

on the segment $[i, i+1]$, at integer m , which is the first term of (39). The structural origin of this gap is the absence of an S -segment in their construction: without the S -modularity-transformed contribution at the $-1/\tau_0$ endpoint in Step 3, the shift-difference equation (37) has only the single right-hand-side term $-\tau_0^{s-1}$, and its solution is the single-Hurwitz kernel. The second $e^{i\pi(s-1)} \tau_0^{k-1-s}$ term appears precisely because the two-segment functional includes the S -segment and its endpoint at $-1/\tau_0$ contributes under τ_0 -differentiation via S -modularity.

B Eichler–Shimura for physicists

This appendix is intended as an exposition aid for readers approaching the subject from theoretical physics. It is removable without affecting the logical content of the paper and is excluded from the journal/arXiv version.

[To be filled in. Topics: cusp forms vs. holomorphic vs. weakly holomorphic vs. meromorphic modular forms $(S_k, M_k, M_k^!, F_k)$; the upper half-plane $\mathbb{H}$ and the modular curve $Y(1) = \Gamma \backslash \mathbb{H}$ as the moduli space $\mathcal{M}_{1,1}$ of elliptic curves; the polynomial coefficient system V_{k-2} and the flat connection on the modular curve; the cocycle / coboundary language of $H^1(\Gamma; V_{k-2})$ and its cuspidal subspace H_{cusp}^1 ; periods and quasi-periods $\omega^\pm, \eta^\pm$ via Eichler integrals; physical analogies (worldsheet, polarization tensors on auxiliary variables, holonomies of flat connections, gauge-invariant observables). Worked computations at $k = 12$ where helpful. $\approx 2\text{--}3$ pages, terse.]

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- [3] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [4] Companion verification scripts at 50-digit mpmath precision. `verify_periods.py`: end-to-end verification of Theorem 1.1 for $f \in \{\Delta, \hat{\Delta}\}$ at $k = 12$, including (a) reproduction of Brown’s $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ from [3] via the finite-contour cocycle pipeline; (b) lockstep across all nine interior monomials $X^{n-(j-1)}Y^{j-1}$ for $j \in \{2, \dots, 10\}$; (c) basepoint independence between $\tau_0 = 0.3 + 1.2i$ and $\tau_0 = -0.2 + 0.9i$; and (d) agreement of the Bernoulli closed-form for δ_T^{-1} in Proposition 3.2 with the LU-decomposition recursion on every coefficient p_m , $m = 0, \dots, 9$. `verify_s6.py`, `verify_multi_s.py`: per-mode verification of Theorem 1.2. Additional scripts for Theorem 1.3 to be added.
- [5] F. Brown, personal communication to the author, 28 June 2019. This correspondence describes the two-segment cocycle procedure (integrate along contours from $\gamma^{-1}\tau_0$ to τ_0 for the two generators $\gamma \in \{S, T\}$ of Γ , modify by a coboundary $P|(1 - T)$ to kill the T -component, then read $\omega^\pm$ and $\eta^\pm$ off the S -component), and is the structural starting point of the construction in the present note. An accompanying unpublished note on quasi-periods of cusp forms of levels 6 and 8 was attached to the correspondence.
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